

# Harsh Kumar

Full Stack & AI/ML Engineer

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## SUMMARY

B.Tech Information Technology graduate (**CGPA 8.03**) with hands-on experience building **full-stack and AI-powered applications** using React, Python, FastAPI, Node.js, REST APIs, databases, and RAG pipelines. Built **3+ end-to-end applications** spanning frontend development, backend APIs, vector search, LLM integration, and real-time systems. Strong foundation in **DSA, OOP, DBMS, OS, and Computer Networks**, with practical experience in generative AI, embeddings, vector databases, and cloud APIs. Seeking **Software Engineering, Full Stack, or AI/ML internship** opportunities.

## EDUCATION

|             |                                                                   |                   |
|-------------|-------------------------------------------------------------------|-------------------|
| 2022 – 2026 | <b>B.Tech in Information Technology, MIT ADT University, Pune</b> | <b>CGPA: 8.03</b> |
| 2022        | <b>Class XII</b> , Kendriya Vidyalaya Southern Command, Pune      | <b>85.8%</b>      |
| 2020        | <b>Class X</b> , Kendriya Vidyalaya Southern Command, Pune        | <b>91.2%</b>      |

## TECHNICAL SKILLS

|                   |                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------|
| <b>Languages</b>  | Python, C++, JavaScript, SQL                                                                        |
| <b>AI/ML</b>      | RAG, LangChain, Embeddings, Vector Search, LLM Integration, Prompt Engineering                      |
| <b>Frameworks</b> | React, React Native, FastAPI, Node.js, Express.js                                                   |
| <b>Databases</b>  | ChromaDB, MySQL, MongoDB, Firebase Realtime Database, Firestore                                     |
| <b>Tools</b>      | Git, GitHub, Postman, AWS, Microsoft Azure, Vite                                                    |
| <b>Core CS</b>    | Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering |

## PROJECTS

### DocuMind – RAG-Powered PDF Assistant

 [GitHub](#)

*React, FastAPI, Python, LangChain, ChromaDB, Hugging Face, Groq, Gemini*

- **Engineered** a full-stack RAG application enabling users to upload **multiple PDFs** and query their contents through a conversational React interface.
- **Built** a 5-stage document pipeline covering PDF extraction, token-based chunking, embedding generation, vector storage in **ChromaDB**, and similarity-based retrieval of the **top-3 relevant chunks**.
- **Developed 6+ FastAPI endpoints** for document processing, semantic search, vector-store inspection, health monitoring, and LLM-powered chat with interchangeable **Groq and Gemini** providers.

### Padh.AI – AI-Powered Academic Learning Platform

 [GitHub](#)

*React, JavaScript, Python, FastAPI, RAG, Embeddings, Supabase*

- **Developed** an AI learning platform that converts academic PDFs into **4 learning outputs**: summaries, concept explanations, PYQ-style questions, and interactive quizzes.
- **Implemented** a modular RAG workflow covering document extraction, preprocessing, chunking, embeddings, similarity retrieval, prompt construction, and LLM-based generation.
- **Integrated** React frontend workflows with **FastAPI and Supabase**, separating document processing, retrieval, generation, and application data layers for scalable feature development.

### Samvad – Real-Time Assistive Communication App

 [GitHub](#)

*React, JavaScript, Firebase, Google Cloud APIs*

- **Developed** a real-time assistive communication app combining **speech-to-text and text-to-speech** to support communication for hearing- and speech-impaired users.
- **Integrated** Google Cloud speech services with **Firebase Realtime Database** to synchronize conversations across devices without manual refreshes.
- **Led a 3-member team** through development and delivery, earning the **Best Departmental Project Award**.

## ACHIEVEMENTS & CERTIFICATIONS

### Best Departmental Project Award

Won for *Samvad*, an assistive communication application – led a **3-member team**.

### Smart India Hackathon 2024

**Top 30 teams** (AI/ML category) – developed an AI-powered solution as part of the university team.

### E-Conjecture Hackathon, IIM Indore

**Finalist**.

### Certifications

AWS Academy Cloud Foundations — Microsoft Azure Cloud Services — SQL (Coursera & IBM) — Python for Data Science (IBM) — Python Essentials 1 (Cisco) — Software Design: Requirements (LinkedIn) — IT Specialist in Databases (Certiport)